

SUPPLEMENTARY NUMERICAL RESULTS FOR PROJECTED MULTILINEAR MODELS

ELIUTH CHAVERO JASSO

ABSTRACT. Nine figures supporting the numerical study [2], together with the information needed to regenerate each of them: the generating script, the input files, the parameters, and checksums of inputs and outputs. Each figure is supplied as a vector PDF, an archival SVG and a raster PNG at 300 dots per inch, from a single generator run. Captions state whether a displayed quantity is an analytical bound, an identity holding by construction, or an observed value, and every logarithmic axis with a plotting floor states that floor. Section 2 lists the data sources, Section 3 the parameters of each experiment, and Section 5 the material that is deliberately not included.

1. FIGURES

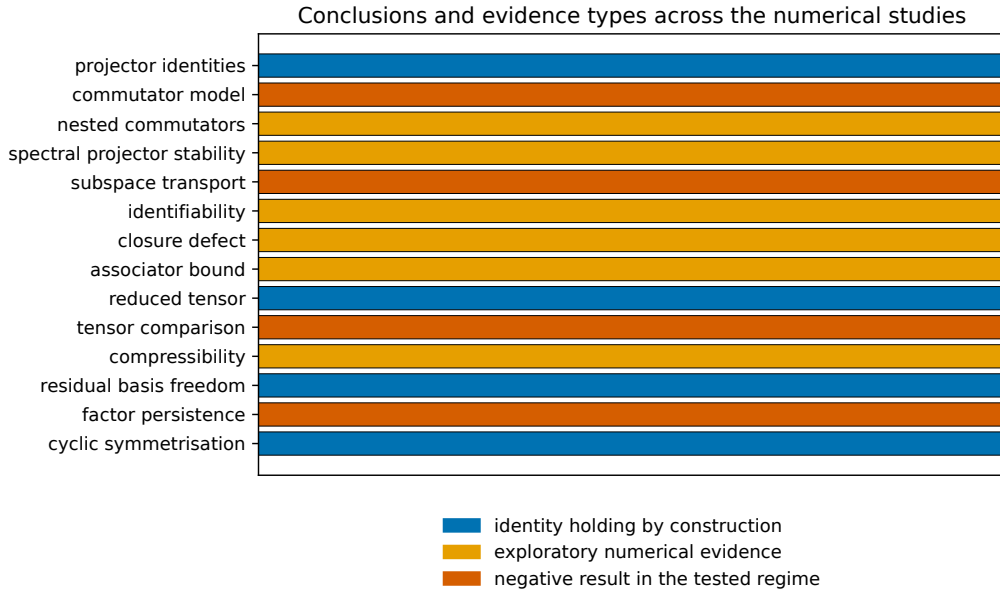


FIGURE 1. Summary of conclusions and evidence types across the numerical studies. Each of the fourteen experiments of [2] is classified into exactly one of three categories: an identity holding by construction, exploratory numerical evidence, or a negative result in the tested regime. The classification is editorial and follows the scheme stated in that article; no experiment contributes to a combined score, and none is ranked against another.

Date: 31 July 2026.

Supplement to [2]. Every figure is regenerated from the committed experiment records by `generate_figures.py`; the mapping from figure to input file, with checksums, is in `figure_provenance.json`.

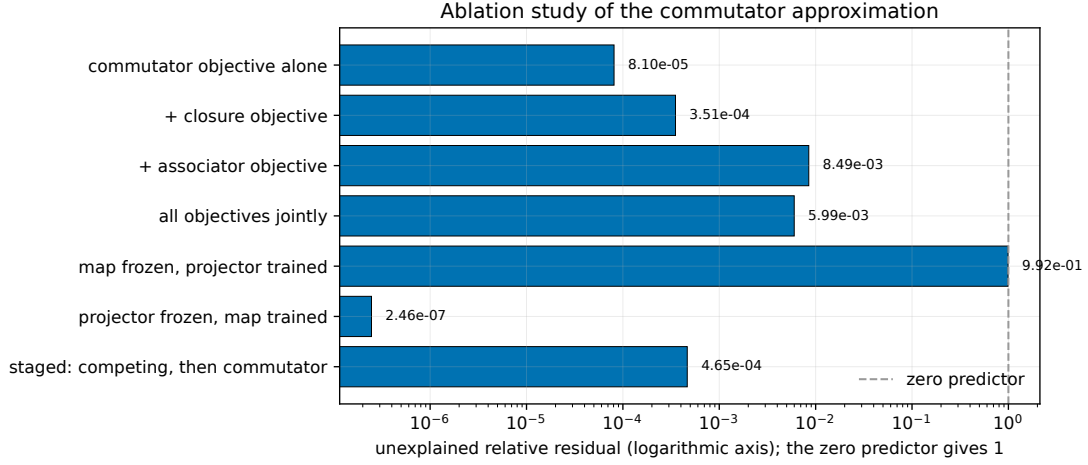


FIGURE 2. Ablation study of the commutator approximation. Seven training regimes, one run each, at ambient dimension 16, reduced rank 4, representation rank 4 and 400 optimisation steps. The horizontal axis is the unexplained relative residual on a logarithmic scale; the dashed line marks the zero predictor, which gives the value 1. Every bar is drawn at the value recorded in the experiment file and none is displayed at a plotting floor; in particular the regime with the projector frozen reaches 2.46×10^{-7} , which is the numerical floor of that run and not zero. Freezing the map and training only the projector leaves the residual at 0.9925; freezing the projector and training only the map, with the projector left at its random initial value, reaches the smallest residual of the seven. One run per regime: this is exploratory evidence, not a replicated experiment.

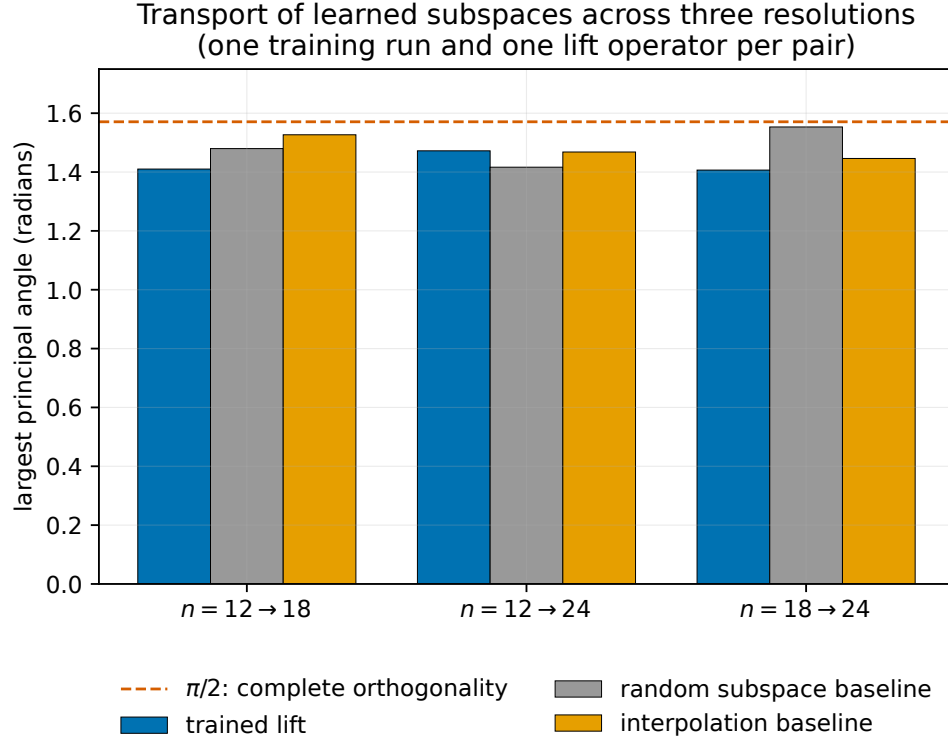


FIGURE 3. Transport of learned subspaces across three independently trained resolutions. For each ordered resolution pair, the largest principal angle between the transported subspace and the independently learned subspace, for the trained lift and for two baselines. The dashed line is $\pi/2 \approx 1.571$ radians, complete orthogonality. All quantities lie between 1.407 and 1.553 radians. The trained lift improves on both baselines in two of the three pairs, by 0.02 to 0.15 radians, which is small relative to how close every condition already is to complete orthogonality. One training run and one lift operator per pair; no uncertainty interval is available and none is drawn.

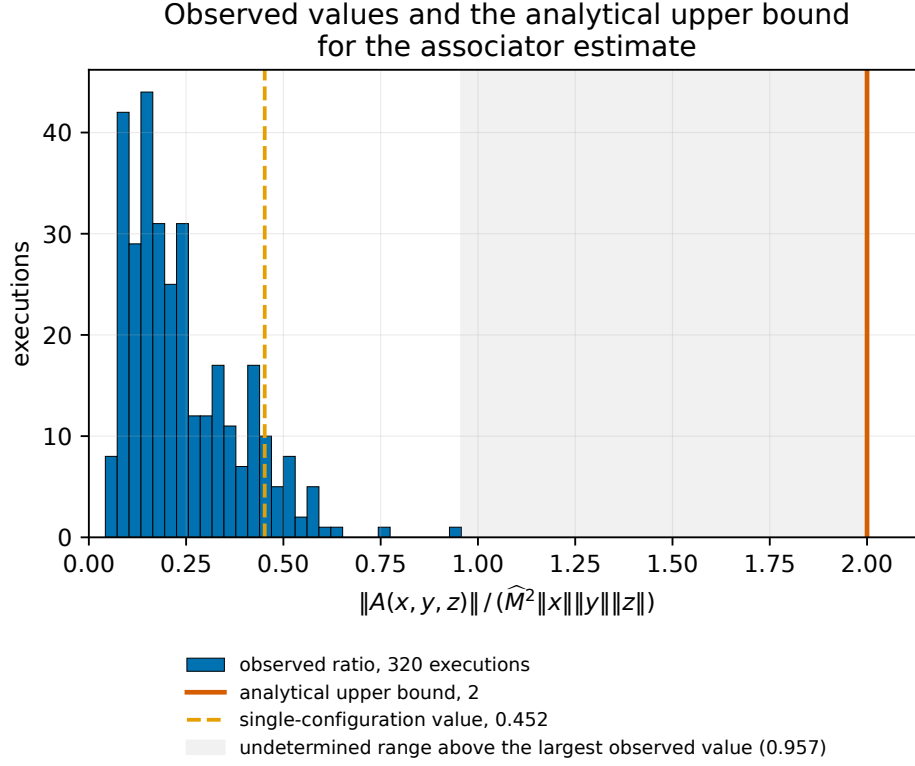


FIGURE 4. Observed values and the analytical upper bound for the associator estimate. Histogram of the observed ratio $\|A(x, y, z)\| / (\widehat{M}^2 \|x\| \|y\| \|z\|)$ over the 320 executions of the broad sweep. The solid vertical line is the triangle bound of value 2, which is an analytical upper bound and not an observation. The dashed line is the single-configuration value 0.452 reported in [2]. The shaded band is the range between the largest observed value and the bound, in which the exact constant is undetermined; it is not an uncertainty interval.

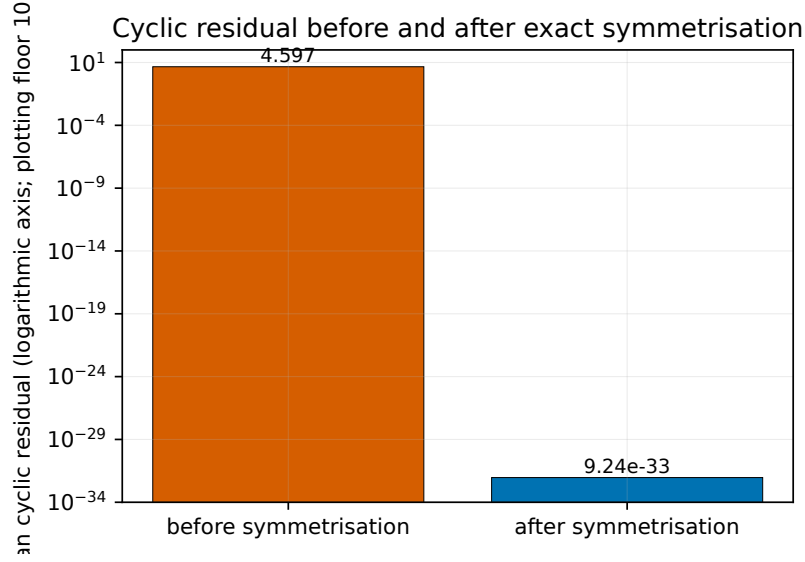


FIGURE 5. Cyclic residual before and after exact symmetrisation, on a logarithmic axis with plotting floor 10^{-34} , as stated on the axis. The value after symmetrisation is at the numerical floor *by construction*: cyclic averaging is an orthogonal projector onto the cyclically invariant subspace, so its output is cyclic identically. The figure confirms that the implementation applies that projector correctly. It is not evidence that a symmetry was learned.

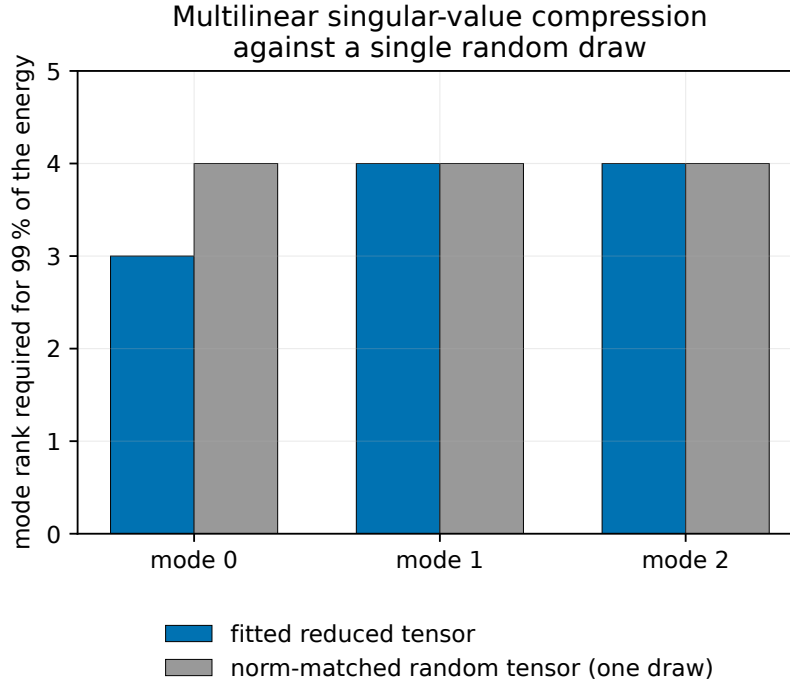


FIGURE 6. Multilinear singular-value compression against a single random draw. Mode ranks required to capture 99 % of the energy, for a fitted reduced tensor and for one norm-matched random tensor, at a single configuration. The fitted tensor is more compressible in one of the three modes and equally compressible in the other two. One random draw is a control, not a null distribution, and no distributional statement follows from this figure.

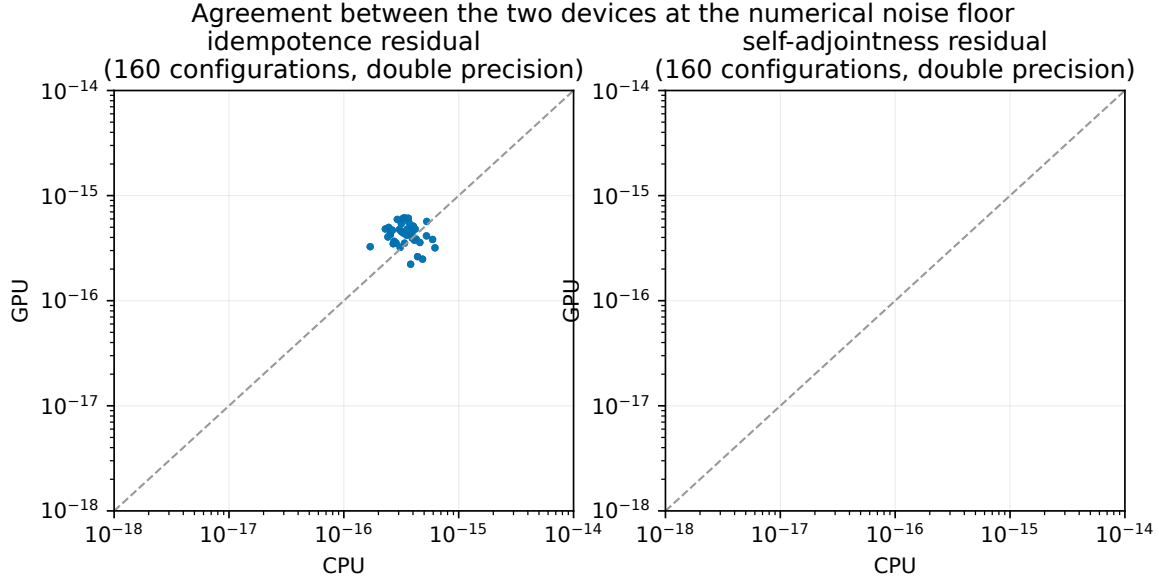


FIGURE 7. Agreement between the two devices at the numerical noise floor. Each point is one configuration, with its CPU value on the horizontal axis and its GPU value on the vertical axis, in double precision with reduced-precision matrix multiplication disabled; the dashed line is equality. Both residuals lie at the noise floor on both devices, so the figure shows agreement of the implementation, not accuracy of any quantity of interest. Values below 10^{-20} are drawn at that floor.

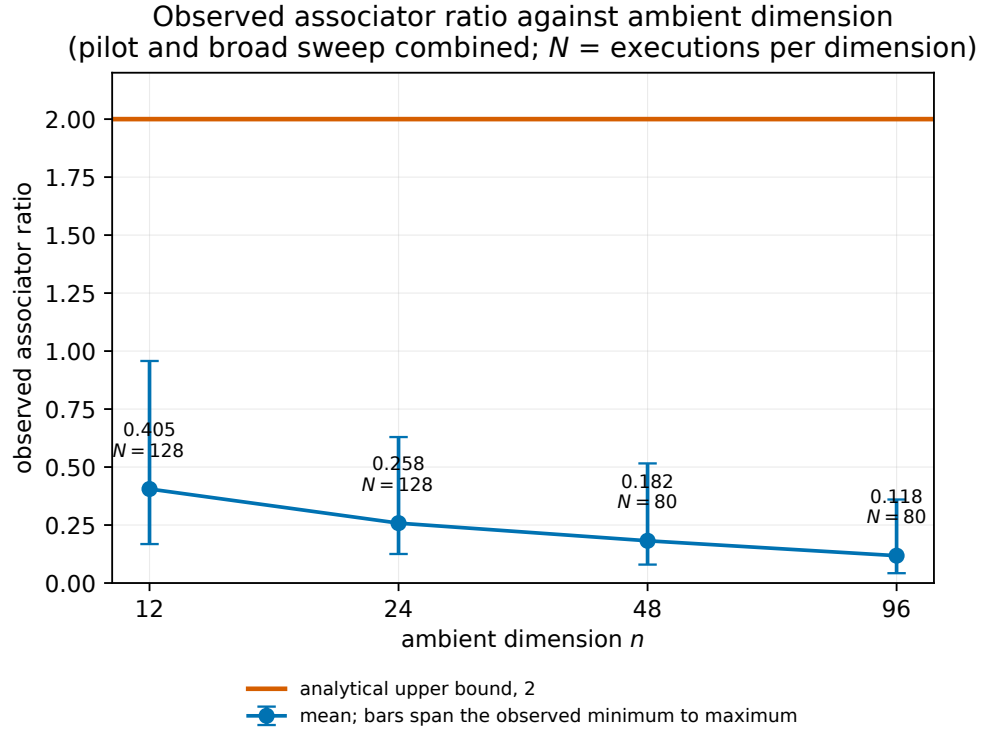


FIGURE 8. Observed associator ratio against ambient dimension, over the pilot and broad sweeps combined. Markers are means and bars span the observed minimum to maximum; the number of executions contributing to each point is annotated. The mean decreases monotonically from 0.405 at $n = 12$ to 0.118 at $n = 96$, moving further from the analytical bound of 2 as the ambient dimension grows, not closer. The single-configuration value 0.452 of Figure 4, measured at $n = 16$, is consistent with this trend. The bars are observed ranges, not confidence intervals. The two executions of a configuration, one per device, are not independent observations; the counts shown are executions.

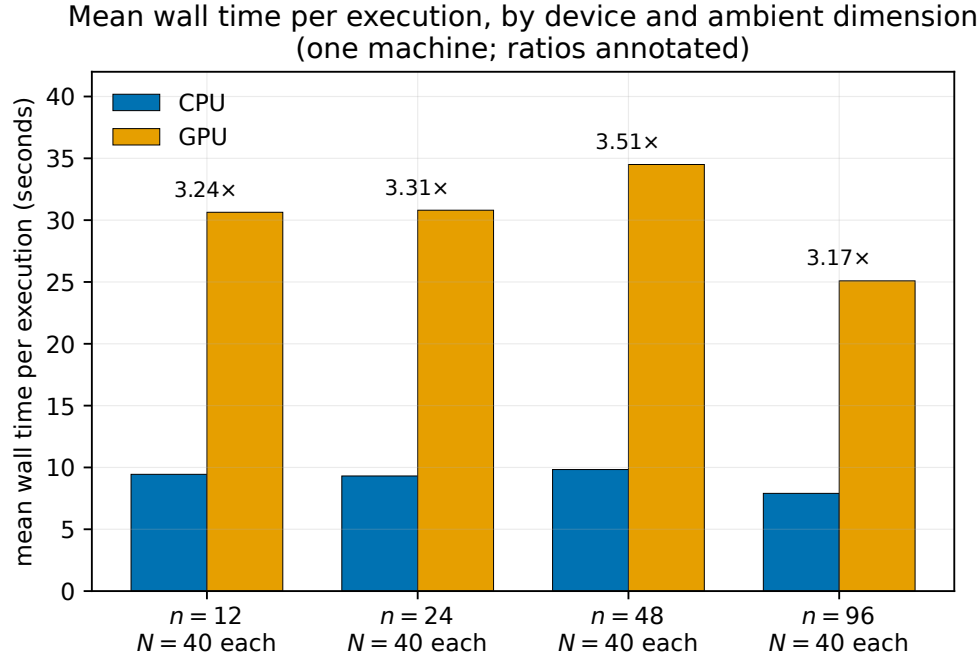


FIGURE 9. Mean wall time per execution, by device and ambient dimension, over the 160 configurations of the broad sweep, with 40 executions per device per dimension. The annotated ratios are between 3.17 and 3.51 and show no trend towards crossover across an eightfold range in ambient dimension. This is a measurement on one machine and one software stack; it supports no asymptotic-complexity statement and no comparison between hardware platforms.

2. DATA SOURCES AND PROVENANCE

Every figure is produced by `generate_figures.py` from files in `data/`, which are verbatim copies of the committed experiment records. The generator writes `figure_provenance.json`, containing for each figure: the input files with their SHA-256 checksums and sizes; the output files in all three formats with their SHA-256 checksums and sizes; the SHA-256 of the generator itself; and the versions of the numerical libraries used. No value in any figure is entered by hand.

TABLE 1. Input file for each figure.

Figure	Input
1	classification stated in [2]; no data file
2	<code>data/block_b_ablation_matrix.json</code>
3	<code>data/block_e_interscale_experiment.json</code>
4	<code>data/s1_results.parquet</code>
5	<code>data/s1_results.parquet</code>
6	mode ranks at the reference configuration, stated in [2]
7	<code>data/s1_results.parquet</code>
8	<code>data/pilot_results.parquet</code> , <code>data/s1_results.parquet</code>
9	<code>data/s1_results.parquet</code>

3. PARAMETERS OF THE SWEPT EXPERIMENTS

TABLE 2. Design of the two sweep stages. A configuration is one distinct mathematical setting; an execution is one run of a configuration on one device. Each configuration was executed once on the CPU and once on the GPU, so the two executions of a configuration are not independent observations.

stage	configurations	executions	grid
pilot	48	96	arity $\in \{3, 4\}$, $n \in \{12, 24\}$, rank $\in \{3, 6\}$, representation rank $\in \{4, 8\}$, 3 seeds
broad	160	320	as above, with $n \in \{12, 24, 48, 96\}$ and 5 seeds
total	208	416	

There were no failures in either stage. Summed over the per-execution records, wall time was 942.8 seconds for the pilot and 6300.8 seconds for the broad stage. Both stages ran in double precision with reduced-precision matrix multiplication explicitly disabled and deterministic algorithms enabled; the hardware is described in [3].

4. VISUAL CONVENTIONS

- The palette is Okabe–Ito, which is colourblind-safe; no encoding relies on distinguishing red from green.
- Every figure is produced as a vector PDF, which is what the manuscript includes; SVG and 300-dpi PNG versions are supplied alongside.
- Where a logarithmic axis cannot display a value, the plotting floor is stated on the axis and the true value is annotated.
- Where a figure shows a range, the caption states whether it is an observed minimum–maximum range or something else. No confidence interval is drawn anywhere in this supplement, because no sampling model is specified for any of these experiments.

- Sample counts are given on the figure or in its caption.
- Optimiser restarts and repeated executions of the same configuration are never used as replicates.

5. WHAT IS DELIBERATELY NOT INCLUDED

The following material is absent, and is listed rather than silently omitted.

- Figures relating to the finite-dimensional theory of projected composition trees — topology, dimension and η dependence, and the geometry of the coefficients k and $k - 1$. These belong to [1] and appear there.
- A full diagram of the residual basis freedom, beyond the single degenerate-eigenspace example described in [2], and an atlas of the permutations and signs of the generalised Jacobi expression. The underlying data exist but were not turned into figures.
- Adversarial maxima of the closure defect across a swept grid. Figure 8 provides the analogous surface for the associator; the closure defect was not swept.
- Device agreement for the ten experiments that were not extended over the grid. Those were run on the CPU only, so no comparison is possible.
- Any figure aggregating the fourteen experiments into a single score. No such score is formed anywhere in this work.

REFERENCES

1. Eliuth Chavero Jasso, *Error propagation under recursive orthogonal projection of finite multilinear composition trees*, 2026, Companion article.
2. ———, *A numerical study of cyclic multilinear maps, orthogonal projectors, and multiresolution tensor representations*, 2026, Companion article, of which this is the supplement.
3. ———, *Software and reproducibility for numerical experiments on projected multilinear models*, 2026, Companion software article.

INDEPENDENT RESEARCHER, APIZACO, TLAXCALA, MEXICO